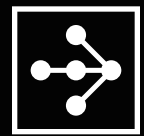


RAY
SUMMIT
2024

Connecting idle GPUs worldwide to fulfill part of the infinite demand for AI Computation

Aline Almeida, IO.net



RAY
SUMMIT
2024

Intro

io.net



Aline Almeida

📍 Germany



aline@io.net

linkedin.com/in/alinealmeida

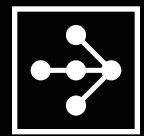


io.net

IOG | FOUNDATION

Partners





- **About IO.net**

- The world's largest decentralized GPU cloud
- "Airbnb" of GPUs

- **Characteristics**

- Decentralized, Heterogeneous, Geographically distributed nodes

- **Measuring Node Reliability**

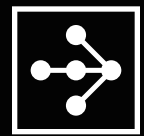
- **Ray.io to Enable GPU Clusters for AI Scaling**

- Customization: Dashboard and Authentication
- Demo - Distributed Training with 56 GPUs

- **Ongoing Optimization Efforts**

- **Research Partnership**





Underutilized Supply meets AI Demand

Independent
Data Centers



Crypto
Mining Farms



Consumer
GPUs



io.net | CLOUD



Ray

Deploy a Decentralized Ray Cluster

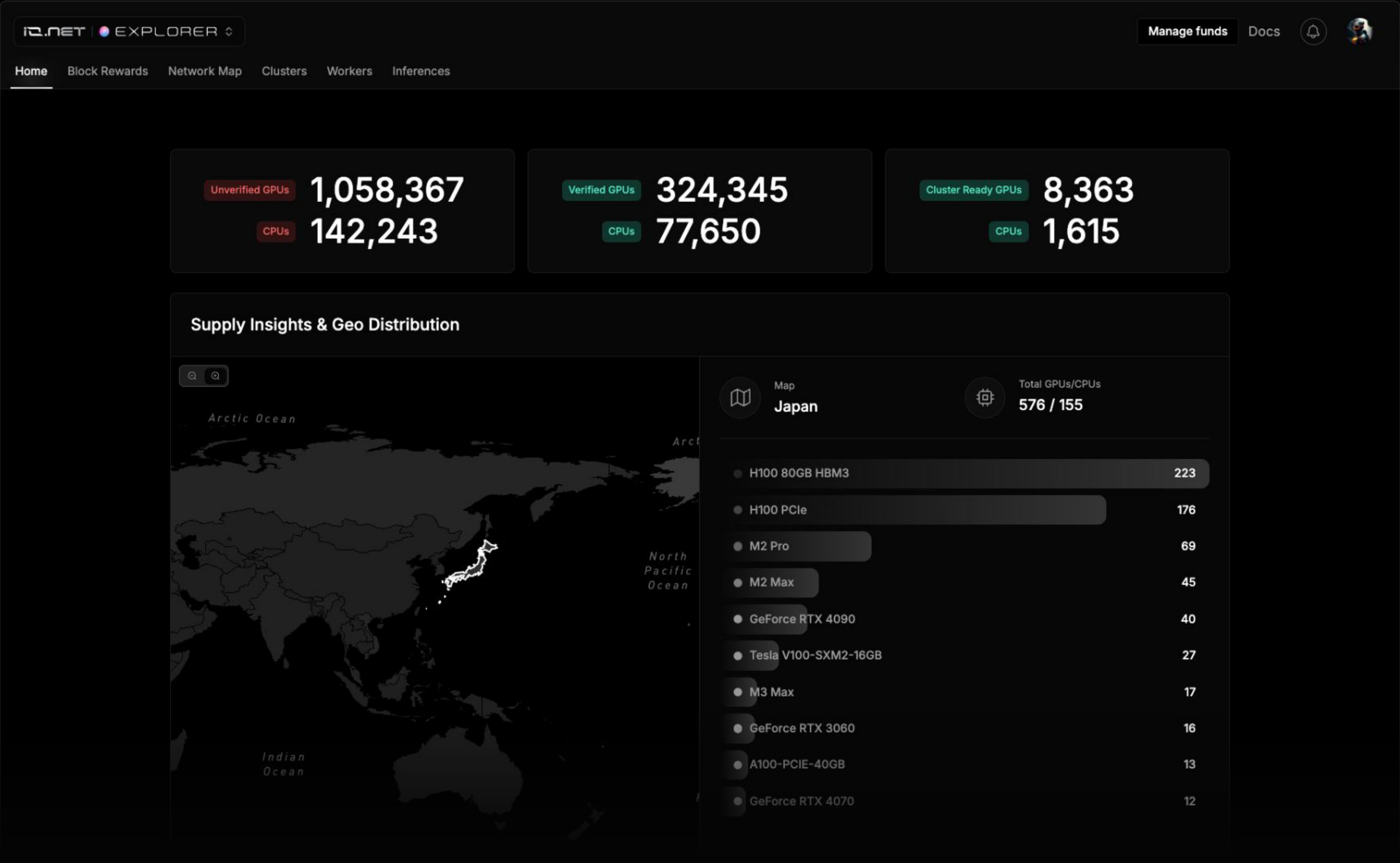


Mega-Ray

Deploy a Decentralized Mega-Ray Cluster

...others







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EXPLORER

Manage funds

Docs

Home

Block Rewards

Network Map

Clusters

Workers

Inferences

Country

Location

US

Washington

Active for: 10 days 01:10:42

Device ID

df73a7...

M2 Pro x1

Active for: 10 days 01:06:38

Device ID

1729e2...

H100 80GB HBM3 x8

Active for: 10 days 01:12:39

Device ID

0e71b8...

GeForce RTX 3090 x1

Active for: 10 days 01:10:39

Device ID

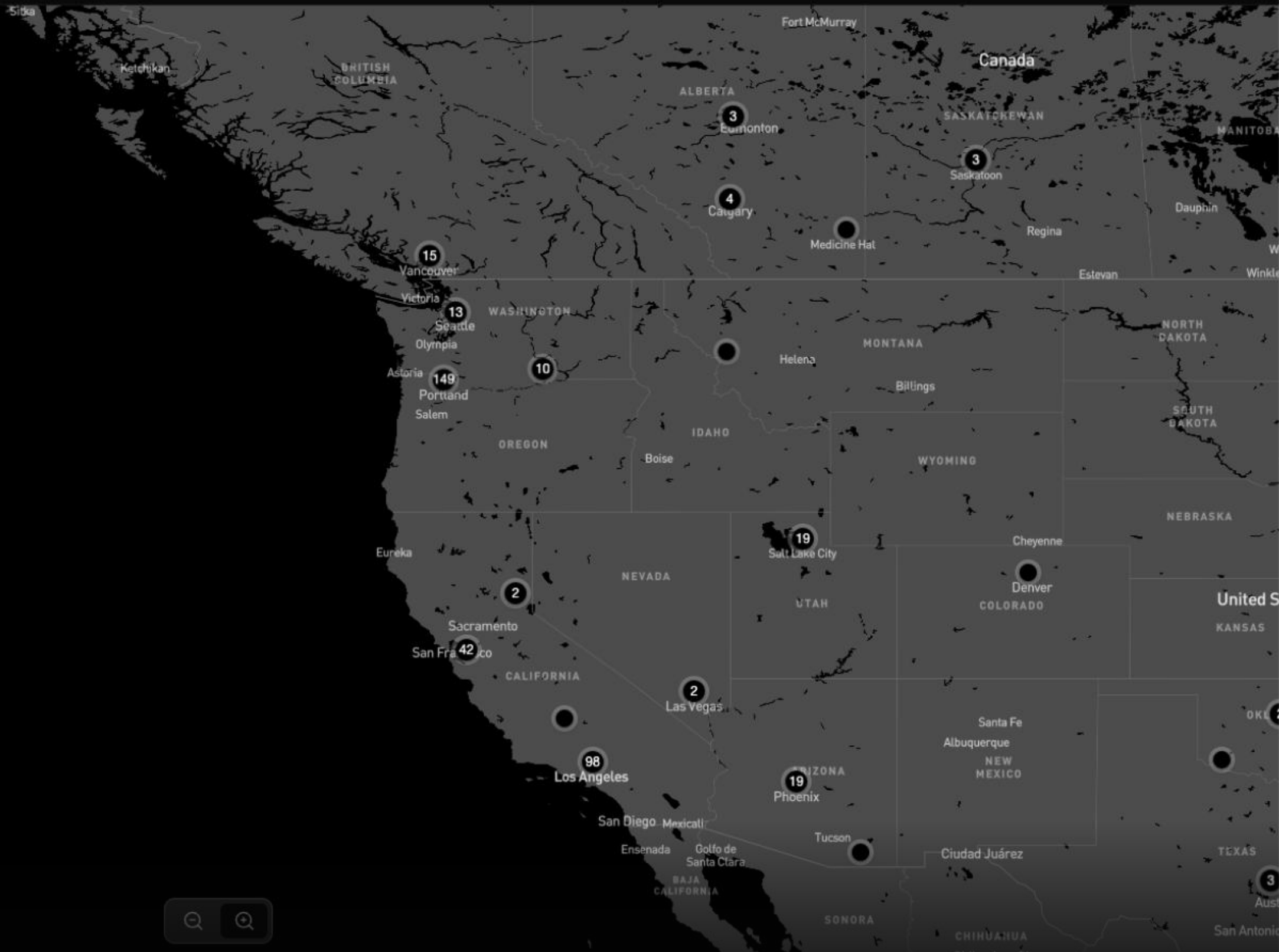
647f2f...

H100 80GB HBM3 x8

Active for: 8 days 14:34:23

1

2

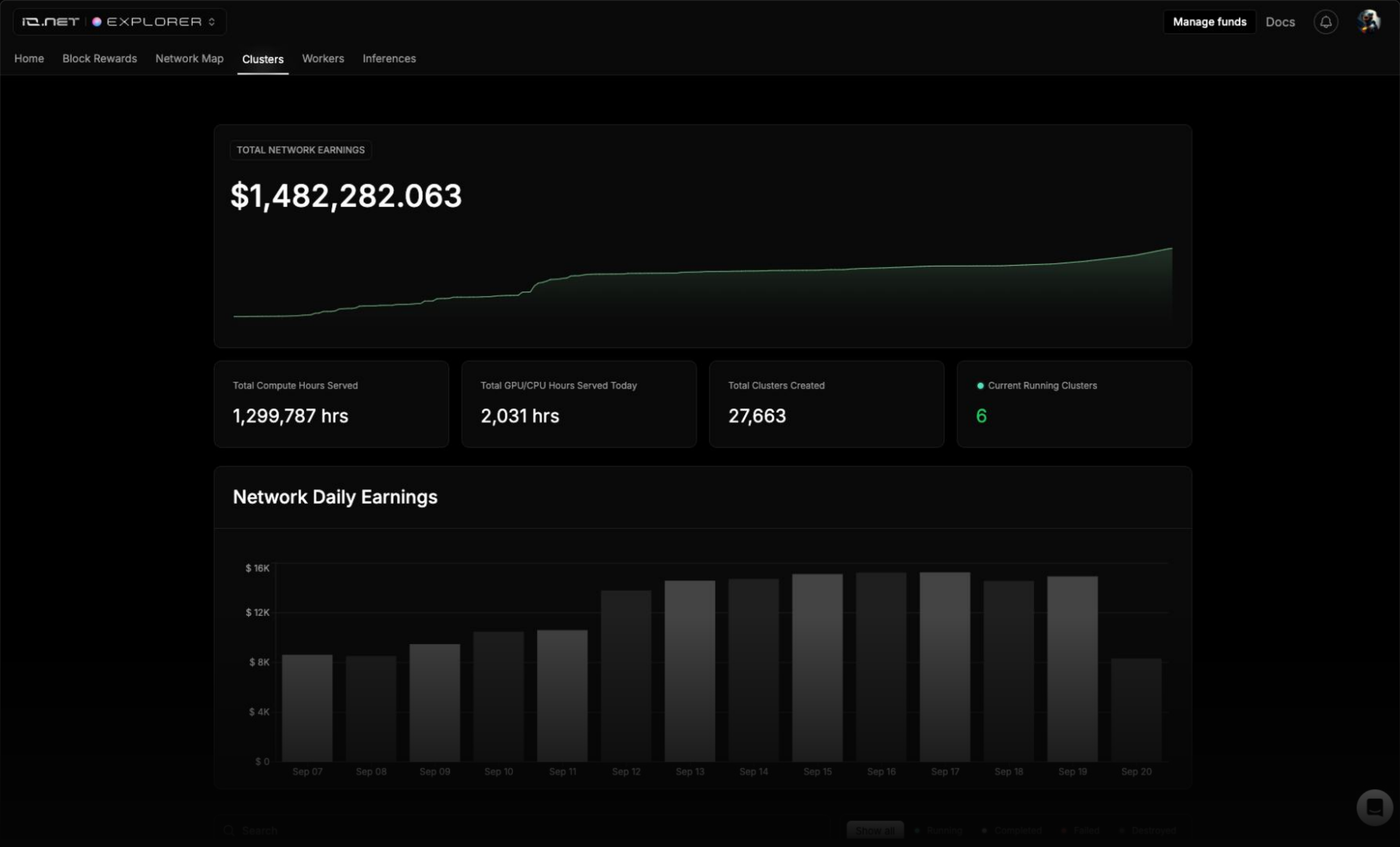
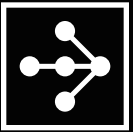


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HomeBlock RewardsNetwork MapClustersWorkersInferences

Total Worker Earnings

\$1,118,628

Today Worker Earnings

\$641

Total Compute Hours Served

1,299,769

io.net

Total GPUs

8153

Total CPUs

1570

Render Network

Total GPUs

438

Total CPUs

46

Filecoin

Total GPUs

1024

Total CPUs

0

Aethir

Total GPUs

828

Total CPUs

0

Search

RunningPausedOfflineTerminatedUnsupportedBlocked

STATUS	DEVICE ID	READINESS	UP FOR	CHIP/GPUS
Running	0119a5fa-936f-4564-a3b1-4ff17f593a8f	Cluster Ready	3 Days 21 Hrs 2 Mins	A10x3
Running	012964f5-f708-4103-819d-a96147b73c2a	Cluster Ready	3 Days 21 Hrs 23 Mins	M2 Prox1
Running	013ff444-7657-4473-b110-76d42fd5a559	Hired	39 Days 13 Hrs 44 Mins	A100 80GB PCIex1
Running	01496a79-c893-444a-9d8e-8503ec89234c	Cluster Ready	1 Days 5 Hrs 40 Mins	A100 80GB PCIex4
Running	014cf06d-988a-4176-9c7e-9ff85aee8c1d	Cluster Ready	3 Days 21 Hrs 28 Mins	GeForce GTX 1080 Ti x6
Running	015ac277-f86e-4a17-9beb-c68a0a9b43df	Cluster Ready	6 Hrs 7 Mins	Tesla V100-SXM2-16GBx6

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< Back to Workers

Device ID: 0107d813-7f33-427a-9f51-132264ee8b77

Cluster ReadyRunningVerified

BLOCK REWARDS NOMINATION ELIGIBILITY STATUS
Eligible

Running For: 3 days 20:55:54Last 30 days (Aug 22 - Sep 20) 98% Uptime

Block Rewards (BR) 1203BR POW (zkTFLOPs Proof) 1405BR Proof of TimeLock 1441Jobs 13

Search by Block IDShow All 1405Completed 1326Failed 79

Proof IDCompleted6b5b822f-e021-0da7-59d0-c33f2c055a0a

Block ID2024-09-20T11:00:00

Date & Time:Sep 20, 2024 11:13:37 UTC

Complexity:90/100

Hash Rate:23,422

Compute Time5 minutes 29 seconds

Proof IDCompleted8ce8c831-72ee-6bb9-c4de-bb7a4c0b2528

Block ID2024-09-20T10:00:00

Date & Time:Sep 20, 2024 10:09:10 UTC

Complexity:90/100

Hash Rate:23,076

Compute Time7 minutes 8 seconds

H100 PCIex8Linux

Uptime Percentage98%

Device ID0107d813-7f33-427a-9f51-132264e...

Lifetime Proof Of Work "zkTFLOPs" ScoreWorker's overall TFLOPs Computed Score1,327 ComputedCompleted

Connectivity TierUltra High Speed841 Mbps258 MbpsDownloadUpload

Security Compliance

io.NET

ID

Manage funds

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Account Settings

Manage Funds

User Acknowledgment

Manage Funds

Manage your funds and see full account transactions

LIFETIME BLOCK REWARDS

0.00

WORKER

Claim Block Rewards

BLOCK REWARDS ALREADY CLAIMED

WORKER

Deploy Worker To Start Earning +

CLOUD USDC BALANCE

2.20

USDC

Add USDC Funds

CLOUD \$IO BALANCE

0.00

CLOUD

Add \$IO Funds

WORKER EARNINGS

0.00

WORKER

Claim Earnings

Deploy Worker To Start Earning +

Transactions

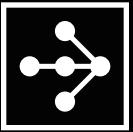
View your transaction history here

Date

Filters

PLATFORM	TYPE OF TRANSACTION	TRANSACTION AMOUNT	DATE	TIME	TRANSACTION ID	
CLOUD	Cluster booking	USDC -2.59	Sep 21, 2024	09:19:09 UTC	7c03d074-71ab-4dfc-9a03-c16512c37dce	...

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RAY SUMMIT 2024

Manage Funds

0.00

20

USD

← Back To Payment Options

Cancel

Confirm Amount

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CLOUD

io.net

COIN

0.00

USD

33.45

Manage funds

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train - Ray App - End-to-End Encrypted - eb86be94

1. Cluster Type

Select the type of nodes aggregated in your cluster.

General

Best for prototyping or general E2E Workloads

Inference

Production ready clusters for low latency inference and heavy workloads

Train

Production ready clusters for machine learning models training and fine tuning

NV Link

SOON

Green GPUs

SOON

Next Step

Creation Steps

Cluster Type

2 Supplier

3 Security Compliance

4 Location

5 Connectivity Tier

6 Select Your Cluster Processor

7 Cluster Base Image

8 Master Configuration

9 Summary

Available GPUs/CPU

9,909

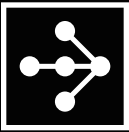
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train - io.net - Ray App - End-to-End Encrypted - eb86be94

Back one step

5. Connectivity Tier

Select the Level of Security For your Cluster. In all choices your model graphs and weights are end to end encrypted and read proof. all the data traffic in-transit between the GPU workers is TLS encrypted.

Next Step

⚡⚡⚡ Ultra High Speed

1.6 Gbps 1.2 Gbps

Download Upload

⚡⚡ High Speed

800 Mbps 600 Mbps

Download Upload

⚡ Medium Speed

400 Mbps 300 Mbps

Download Upload

🏠 Low Speed

100 Mbps 10 Mbps

Download Upload

Creation Steps

✓ Cluster Type

✓ Supplier

✓ Security Compliance

✓ Location

✓ Connectivity Tier

6 Select Your Cluster Processor

7 Cluster Base Image

8 Master Configuration

9 Summary

Available GPUs/CPU

9,403

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< Back one step

6. Select Your Cluster Processor

Browse from a wide range of Chips suitable for different use cases, You Can Either Select a CPU Only Cluster or a GPU Only Cluster

Next Step

Apple

NVIDIA

GPU

Q Search

H100 80GB HBM3

322

A100-SXM4-80GB

202

GeForce RTX 3090

109

H100 PCIe

75

A100 80GB PCIe

66

Creation Steps

✓

Cluster Type

✓

Supplier

✓

Security Compliance

✓

Location

✓

Connectivity Tier

Select Your Cluster Processor

7

Cluster Base Image

8

Master Configuration

9

Summary

Available GPUs/CPUs

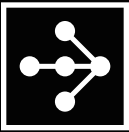
3,141

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Enter GPUs Quantity

Quantity8

Enter Duration

Note: Bookings are not permitted under 1hr

HourlyDailyWeekly

1

io COIN

Cloud Services Fee: 0%

USDC

Cloud Services Fee: 2%

Quantity (GPUs/CPUs)

8

Average Per Card

1.580 USDC/hr

Hours

1 hour

Total Cost Per Card

1.580 USDC

Cost

12.640 USDC

Cloud Services Fee

0.253 USDC

IONET FEE (0.25%)

0.0316 USDC

1 Location Selected

Germany

Cluster Summary

Processor NVIDIA A100-SXM4-80GB

Processor Quantity 8

Connectivity Ultra High Speed

Billing Duration 1 Hour

Cluster Purpose Image Ray App

Security Compliance End-to-End Encrypted

Total Cost

USDC 12.92

Deploy Cluster

✓

Security Compliance

✓

Location

✓

Connectivity Tier

✓

Select Your Cluster Processor

✓

Cluster Base Image

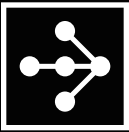
✓

Master Configuration

Summary

Available GPUs

68



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A100-SXM4-80GB - train - io.net - Ultra High Speed - Ray App - End-to-End Encrypted - 3be5542c

Deployment

Deployment started less than a minute ago

Session will expire in: 19m 16s

Cluster deployment can take up to 5 minutes

Payment Verified

Deployment Requested

Deploying

Encrypting Cluster Network & Assigning Access Keys

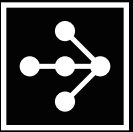
Running Checks

Completed

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A100-SXM4-80GB - train - io.net - Ultra High Speed - Ray App - End-to-End Encrypted - 3be5542c 

Deployment 

Deployment started less than a minute ago

View Cluster

 Payment Verified

 Deployment Requested

 Deploying

 Encrypting Cluster Network & Assigning Access Keys

 Running Checks

 Completed

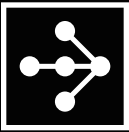
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A100-SXM4-80GB - train - io.net - Ultra High Speed - Ray App - End-to-End Encrypted - 3be5542c

Extend Cluster

Running For: 0 Hrs 15 Mins

Visual Studio

Jupyter Notebook

Ray Dashboard

While accessing to the Ray dashboard, use your IDE password as both the username and password if required.

All Workers

2 Workers

16 GPUs

Search

Running

IO Worker 1

Uptime in Cluster: 100%

Device ID8390cd4d-ca6d-436d-8576-008e28cff0ecA100-SXM4-80GBx8

Running

IO Worker 2

Uptime in Cluster: 100%

Device ID5f5d58bc-8c35-4fb1-a5ae-fe529861bb79A100-SXM4-80GBx8

Cluster ID

A100-SXM4-80GBx16Ray App

5937fa5f-71f0-476f-b3c2-e02d66524e0cCopy

Compute Hours

Running

Served:0 Hrs 15 MinsRemaining:0 Hrs 45 Mins

CLUSTER STARTEDSep 20, 2024 14:00:55 UTCCLUSTER ENDED---

PAID25.85 USDCREFUNDED0.00 USDC

Connectivity Tier

Ultra High Speed

1.6 GbpsDownload1.2 GbpsUpload

Security Compliance

End-to-End Encrypted

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Tesla V100-SXM2-16GB - train - io.net - Medium Speed - Ray App - End-to-End Encrypted - 4e99a41d

IDE PasswordgfeIStW#O...

Extend ClusterRunning For: 0 Hrs 2 Mins

Visual StudioJupyter NotebookRay Dashboard

While accessing to the Ray dashboard, use your IDE password as both the username and password if required.

All Workers10 Workers56 GPUsSearch

Running

IO Worker 1

Uptime in Cluster: 100%

Device ID21b93c7f-f111-40e4-8197-44ab95c054bfTesla V100-SXM2-16GBx6

Running

IO Worker 2

Uptime in Cluster: 100%

Device IDad3cf5bb-7b81-4efd-a726-27d4f849f528Tesla V100-SXM2-16GBx5

Running

IO Worker 3

Uptime in Cluster: 100%

Cluster IDTesla V100-SXM2-16GBx56Ray App

ae4dde4a-33e2-4c09-87a9-a33f139f39fbCopy

Compute HoursRunning

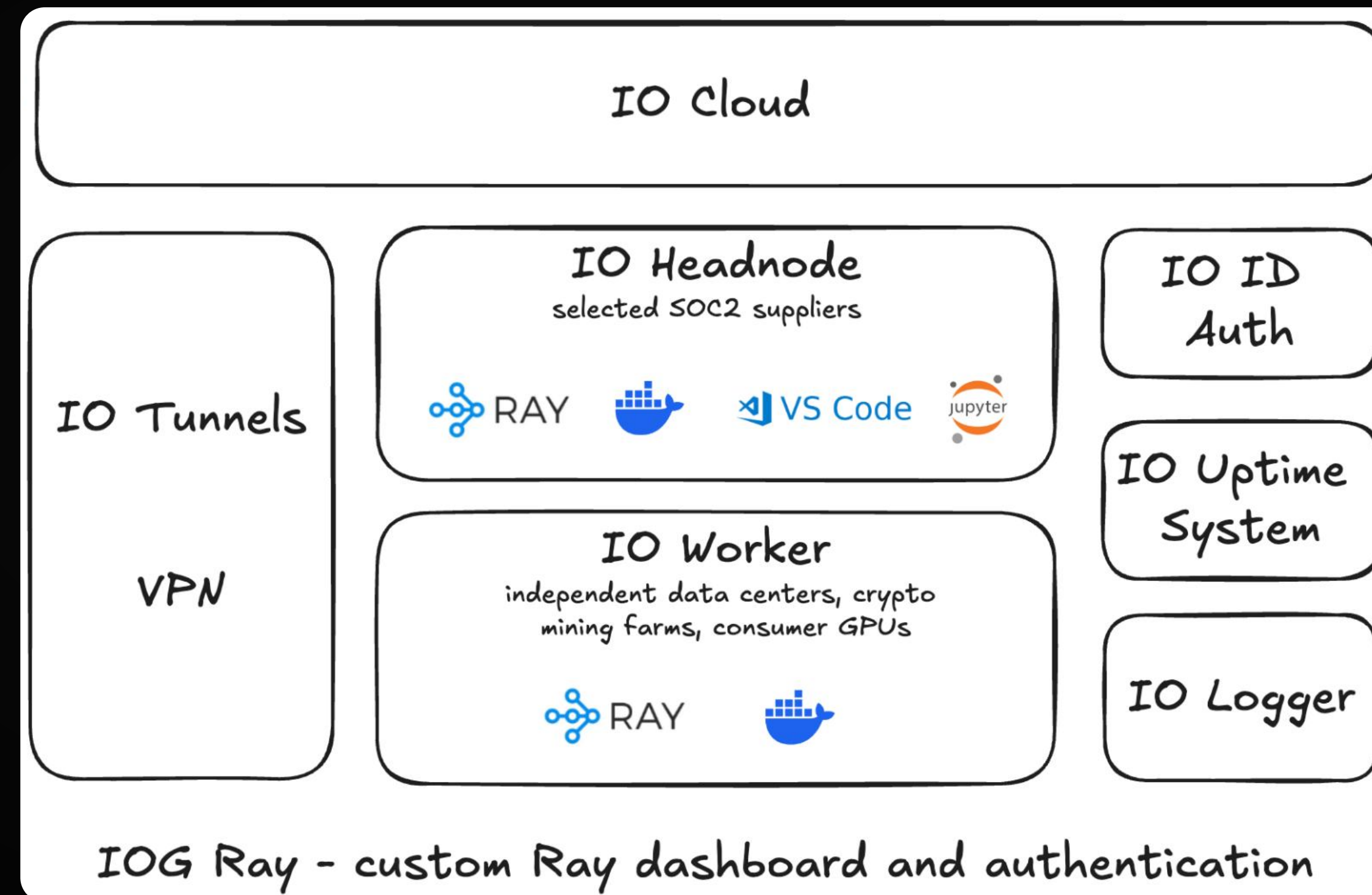
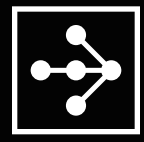
Served:0 Hrs 2 MinsRemaining:0 Hrs 58 Mins

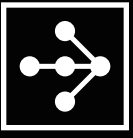
CLUSTER STARTEDSep 22, 2024 22:38:29 UTCCLUSTER ENDED---

PAID16.04 USDCREFUNDED0.00 USDC

Connectivity Tier

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jupyter ray-summit-demo Last Checkpoint: 57 seconds ago

File Edit View Run Kernel Settings Help Trusted

+ - ✂ 📄 📄 ▶ ■ ↺ ▶▶ Code ▾

Open in... Python 3 (ipykernel) ○

[1]: `import ray`

2024-09-22 15:42:25,331 INFO util.py:154 -- Outdated packages:
ipywidgets==7.7.2 found, needs ipywidgets>=8
Run 'pip install -U ipywidgets', then restart the notebook server for rich notebook output.

[2]: `ray.init()`

2024-09-22 15:42:26,968 INFO worker.py:1585 -- Connecting to existing Ray cluster at address: 100.115.112.62:6379...
2024-09-22 15:42:26,980 INFO worker.py:1761 -- Connected to Ray cluster. View the dashboard at **127.0.0.1:8265**

[2]:

 RAY

Python version: 3.9.19

Ray version: 2.30.0

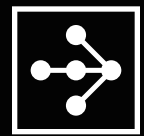
Dashboard: <http://127.0.0.1:8265>

[4]: `ray.cluster_resources()`

[4]:

```
{'GPU': 56.0,  
'node:100.112.169.99': 1.0,  
'CPU': 390.0,  
'memory': 134938159928.0,  
'object_store_memory': 59258226274.0,  
'accelerator_type:V100': 10.0,  
'node:100.76.72.113': 1.0,  
'node:100.121.31.109': 1.0,  
'node:100.115.112.62': 1.0,  
'node:__internal_head__': 1.0,  
'node:100.64.107.8': 1.0,  
'node:100.119.181.63': 1.0,  
'node:100.116.94.59': 1.0,  
'node:100.119.201.125': 1.0,  
'node:100.74.43.22': 1.0,  
'node:100.90.103.34': 1.0,  
'node:100.68.238.73': 1.0}
```

[]:



RAY
SUMMIT
2024

Distributed Training - Example with 56 GPUs

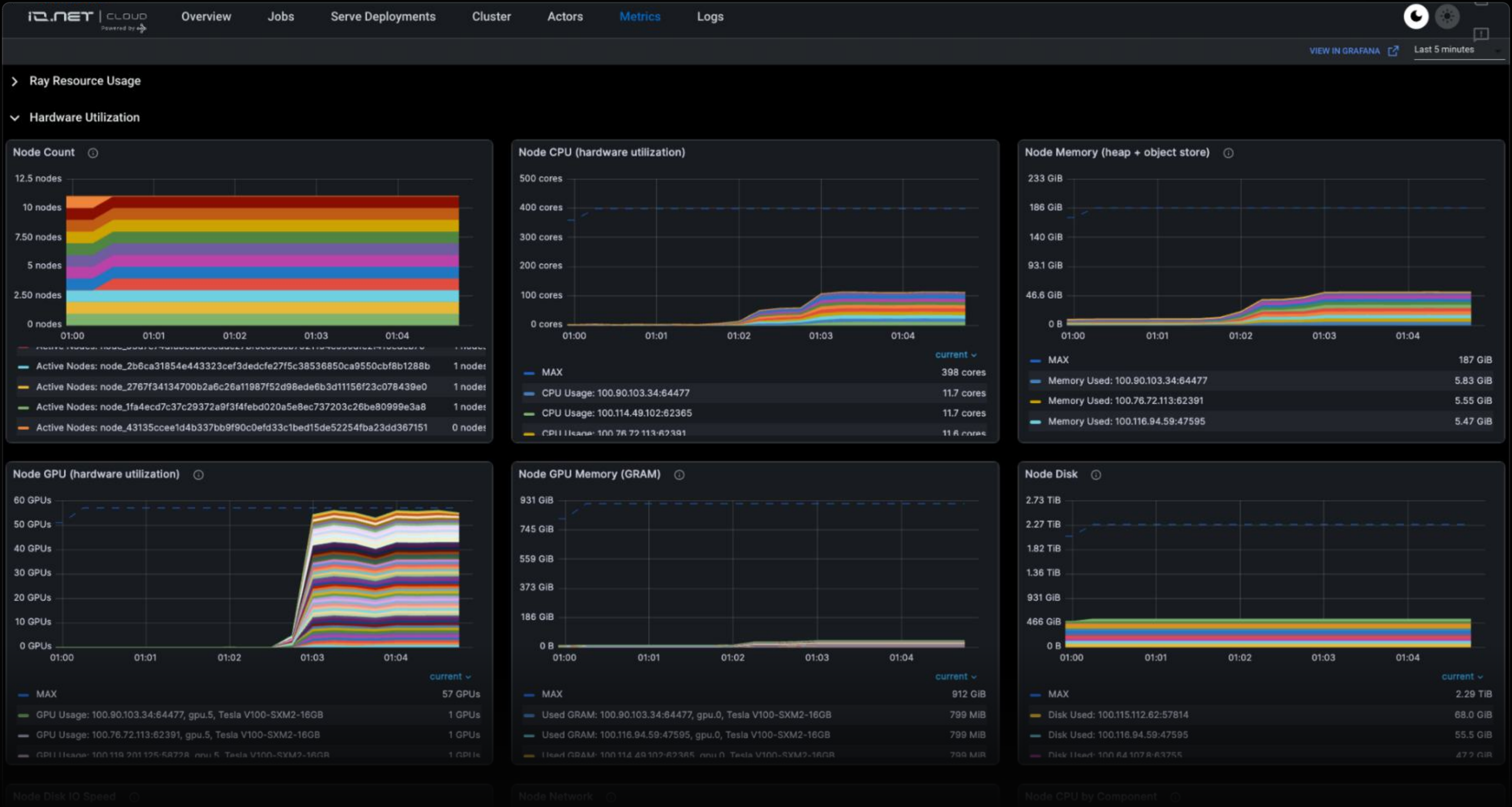
io.net

The screenshot displays a code editor interface with a dark theme. On the left, the 'EXPLORER' sidebar shows a file tree with 'SAMPLES' expanded, containing 'train.py'. The main editor area shows the 'train.py' file, which is a Python script for distributed training using Ray. The script includes imports for 'os', 'torch', 'torchvision', 'ray', and 'torch.nn'. It defines a 'TorchTrainer' class and a 'main' function. The 'main' function sets up a Ray training job with 56 GPUs. The bottom panel shows the 'TERMINAL' output, which displays the Ray training job's progress, including the number of workers, the world rank, and the local rank for each worker. The output shows that the job is running successfully on 56 GPUs.

```
1 # Adapted from
2 # https://docs.ray.io/en/latest/train/examples/pytorch/torch_fashion_mnist_example.html
3 import os
4 from typing import Dict
5 import torch
6 from filelock import FileLock
7 from torch import nn
8 from torch.utils.data import DataLoader
9 from torchvision import datasets, transforms
10 from torchvision.transforms import Normalize, ToTensor
11 from tqdm import tqdm
12
13 import ray.train
14 from ray.train import ScalingConfig
15 from ray.train.torch import TorchTrainer
16
17 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.116.94.59, pid=744) world_rank=0, local_rank=0, node_rank=0
18 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.116.94.59, pid=748) world_rank=1, local_rank=1, node_rank=0
19 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.116.94.59, pid=772) world_rank=2, local_rank=2, node_rank=0
20 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.116.94.59, pid=776) world_rank=3, local_rank=3, node_rank=0
21 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.116.94.59, pid=803) world_rank=4, local_rank=4, node_rank=0
22 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.116.94.59, pid=859) world_rank=5, local_rank=5, node_rank=0
23 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.119.181.63, pid=743) world_rank=6, local_rank=0, node_rank=1
24 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.119.181.63, pid=744) world_rank=7, local_rank=1, node_rank=1
25 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.119.181.63, pid=748) world_rank=8, local_rank=2, node_rank=1
26 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.119.181.63, pid=775) world_rank=9, local_rank=3, node_rank=1
27 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.119.181.63, pid=909) world_rank=10, local_rank=4, node_rank=1
28 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.119.181.63, pid=995) world_rank=11, local_rank=5, node_rank=1
29 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.121.31.109, pid=743) world_rank=12, local_rank=0, node_rank=2
30 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.121.31.109, pid=744) world_rank=13, local_rank=1, node_rank=2
31 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.121.31.109, pid=748) world_rank=14, local_rank=2, node_rank=2
32 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.121.31.109, pid=801) world_rank=15, local_rank=3, node_rank=2
33 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.121.31.109, pid=825) world_rank=16, local_rank=4, node_rank=2
34 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.74.43.22, pid=871) world_rank=17, local_rank=0, node_rank=3
35 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.74.43.22, pid=898) world_rank=18, local_rank=1, node_rank=3
36 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.74.43.22, pid=902) world_rank=19, local_rank=2, node_rank=3
37 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.74.43.22, pid=981) world_rank=20, local_rank=3, node_rank=3
38 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.74.43.22, pid=1146) world_rank=21, local_rank=4, node_rank=3
39 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.76.72.113, pid=742) world_rank=22, local_rank=0, node_rank=4
40 (TorchTrainer pid=651, ip=100.116.94.59) - (ip=100.76.72.113, pid=743) world_rank=23, local_rank=1, node_rank=4
```



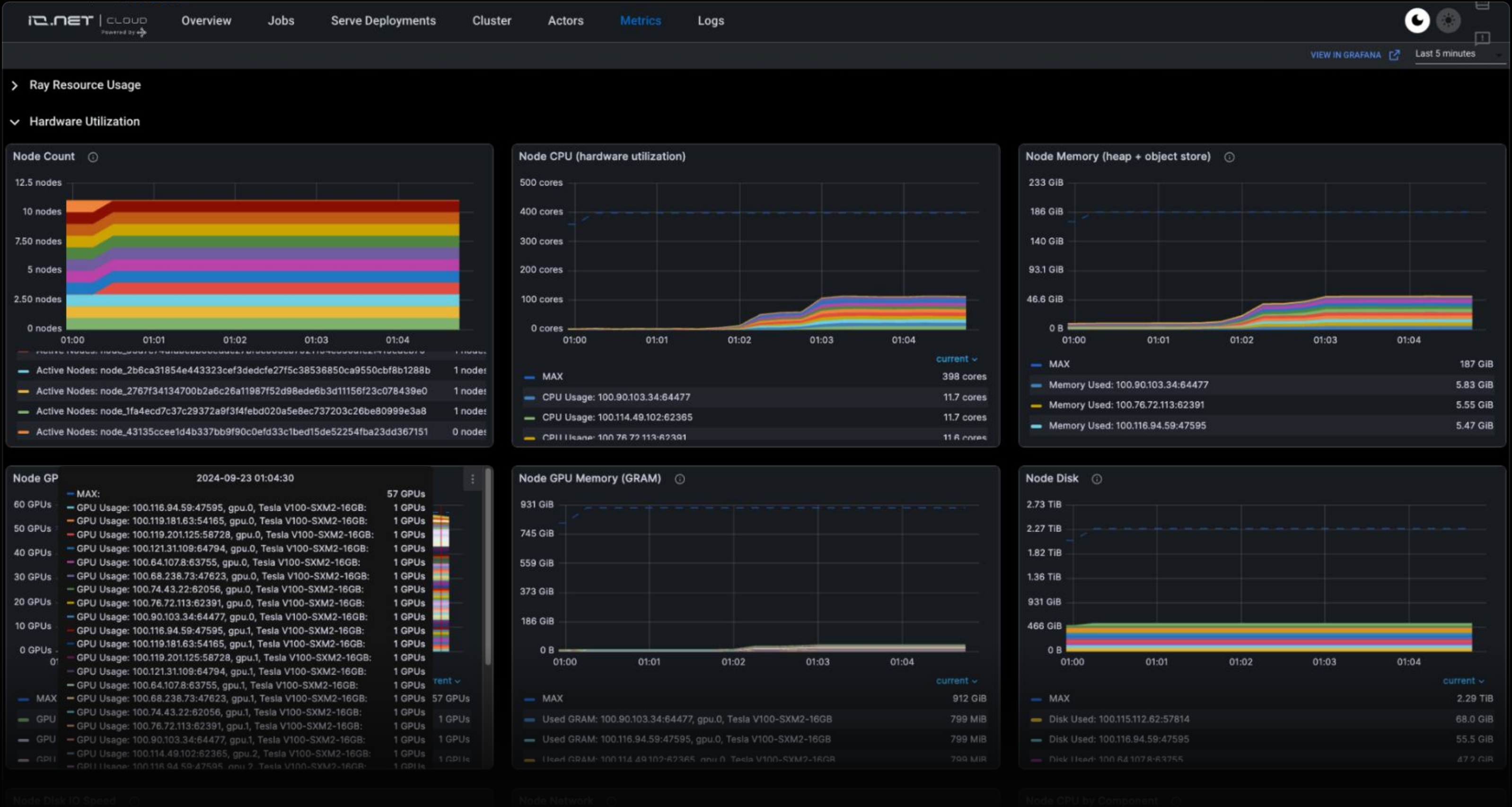

Distributed Training - Example with 56 GPUs





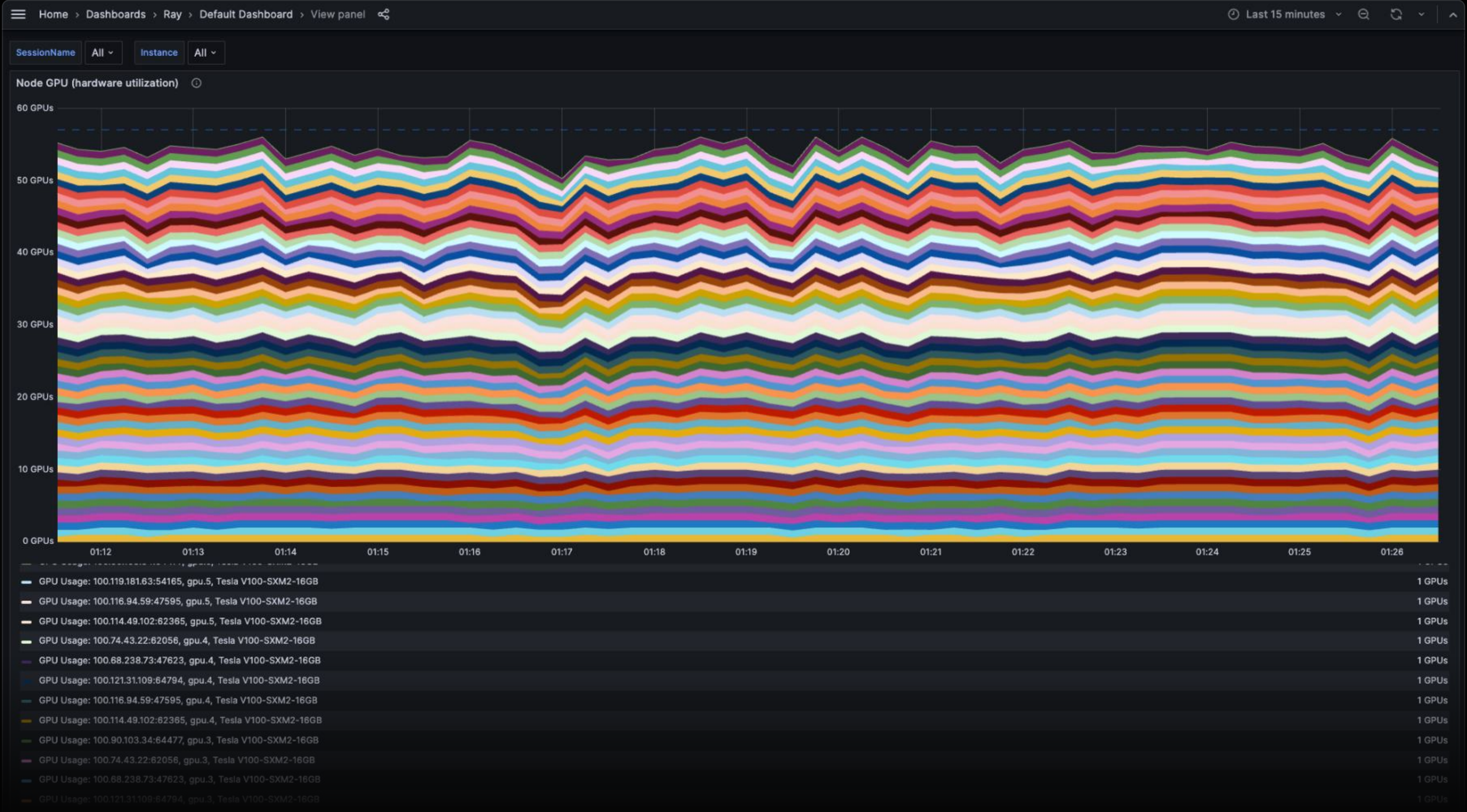
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Distributed Training - Example with 56 GPUs



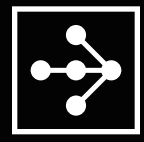


Distributed Training - Example with 56 GPUs





- **IOG Foundation - Developer Tools**
- **Given a cluster request, location, type of workload, ...**
 - Select the subset of nodes that satisfy conditions, besides reliability score
 - Maximum edge of the fully-connected subgraph (clique) is under the maximum latency allowed for the type of AI workload
- **Minimize Communication Overhead**
 - Be opinionated on trade-offs of more nodes vs communication overhead
 - Given a running cluster
 - Optimize resource allocation and task scheduling
 - Optimize topology / communication protocol for custom use cases
 - A fully-connected graph / network is not always needed



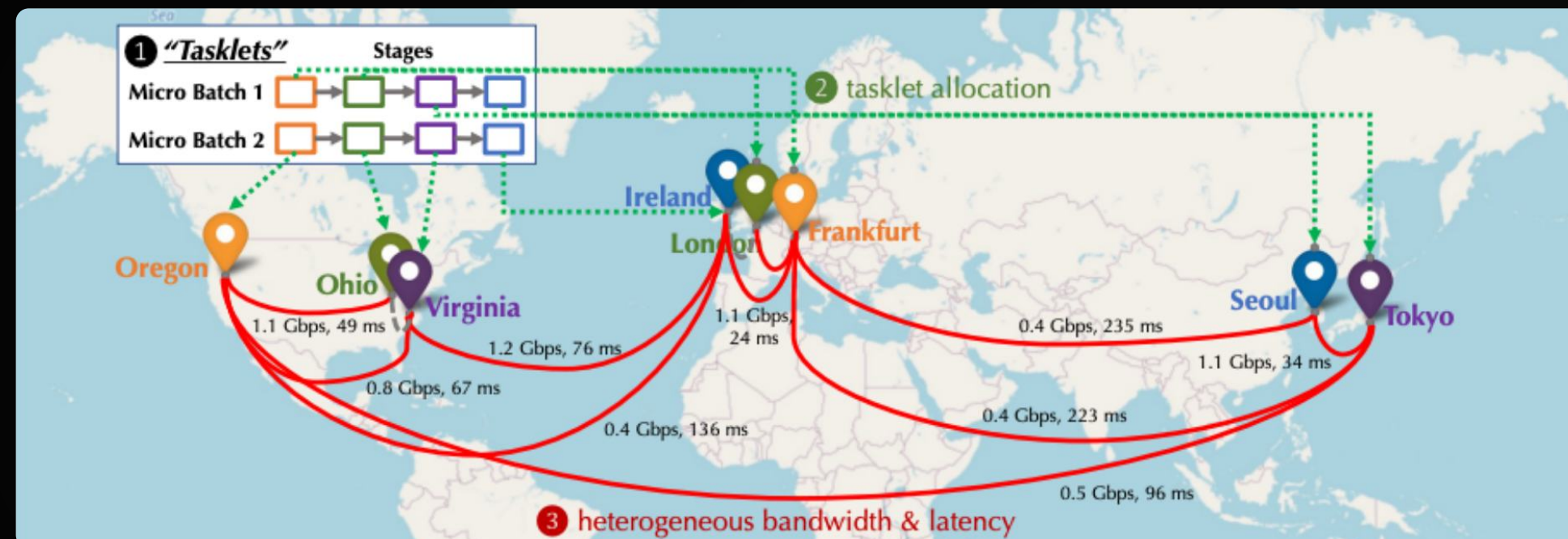
- **Open for Research Partnerships aligned with Decentralized Training / Inference, such as:**

Distributed Inference and Fine-tuning of Large Language Models Over The Internet

37th Conference on Neural Information Processing Systems (NeurIPS Dec 2023)

Alexander Borzunov - HSE University, Yandex & Max Ryabinin - HSE University, Yandex & Artem Chumachenko Neuro.ai Dmitry Baranchuk Yandex & Tim Dettmers University of Washington & Younes Belkada Hugging Face Pavel Samygin Yandex School of Data Analysis & Colin Raffel Hugging Face

<https://arxiv.org/abs/2312.08361>



Decentralized Training of Foundation Models in Heterogeneous Environments

Conference on Neural Information Processing Systems (NeurIPS Jun 2023).

Binhang Yuan^{1*}, Yongjun He^{1*}, Jared Quincy Davis², Tianyi Zhang², Tri Dao², Beidi Chen^{3,4}, Percy Liang², Christopher Re², Ce Zhang¹ ¹ETH Zürich, Switzerland ²Stanford University, USA ³Carnegie Mellon University ⁴Meta AI

<https://arxiv.org/abs/2206.01288>

- **Make research reproducible**

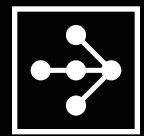
Provide blueprint for researchers to test and run novel algorithms in the field



- **IOG Foundation sponsors Research with:**
 - Infrastructure for testing and validating novel decentralized large-scale distributed training and inference
 - Partnerships with leading academic and industrial researchers to foster innovation in AI scalability

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